

CS 305 Lab Tutorial

Lab9 Router, Cloud in eNSP

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Topic

- Part A. Router
 - Interface of Route
 - features and configuration
 - Route and forward
 - Routing-table
 - direct routing entry
 - static routing entry
 - Other functions of the router
 - DHCP service
- Part B. Cloud in eNSP
 - interconnection(1)

Tips on eNSP

- useful commands while in CLI
 - display
 - dis this
 - dis int brief
 - dis ip routing-table
 - configure interface
 - ip address [IPv4-address in dotted decimal] [mask]
 - configure static routing entry
 - ip route-static [destination-netID] [mask] [next-hop]
 - Configure dhcp
 - dhcp enable
 - ip pool
 - dhcp select ...

Part A. Router

A **router** is a hardware device that **connects two or more networks** and serves as a gateway between them. It is a specialized intelligent network device that **reads the address** in each packet and **determines how to transmit it**.

Function list:

- connects two or more networks
 - different types of the interfaces
 - transmit packet by the packet's destination and the route table
- communicate with other routers
 - Routing protocol
- other function:
 - telnet server, dhcp server,...etc.

the front



the back



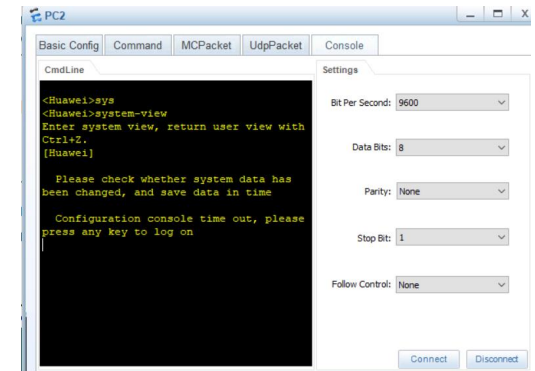
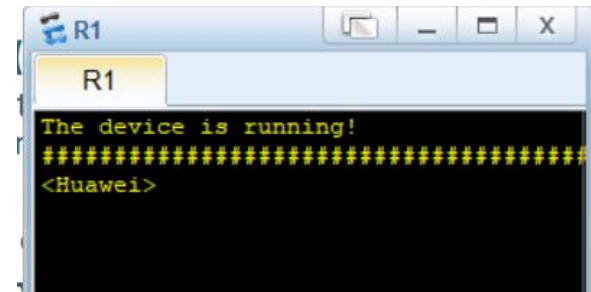
Router: in eNSP

Router (Universal router)

- Function list: IP application(IPv4/IPv6,DHCP,NAT,IPv6), IP routing, Multicast .etc.

How to configure a router on eNSP

- Double click on the Router to invoke the **CLI**(Command Line Interface)
- **PC** connect with Router by **CTL**(Console Line),PC logs in to the router through the terminal and **configures the router in the terminal window**



NOTE: in eNSP, Network devices can only be set up after being started!

Router: interface(1)

1. interface

➤ Function

- Ethernet(for connection), Console (for configuration by CLI)

➤ Bandwidth

- Ethernet(10Mbps), Fastethernet(100Mbps), Gigaethernet(1000Mbps)

➤ Layer

- Layer2(no IP function), Layer3(IP function)
- Layer2/3(Can switch between Layer2 and Layer3)



Router: interface(2)

2. Commands *in system view*

- `display interface port-name port-index`

```
[Huawei]dis int Ethernet 0/0/0
Ethernet0/0/0 current state : UP
Line protocol current state : UP
Last line protocol up time : 2023-11-18 20:30:45 UTC-08:00
Description:
Route Port,The Maximum Transmit Unit is 1500
Internet Address is 172.168.1.1/24
IP Sending Frames' Format is PKTFMT_ETHNT_2, Hardware address is 5489-9877-2d2a
Last physical up time : 2023-11-18 20:30:45 UTC-08:00
Last physical down time : 2023-11-18 20:30:42 UTC-08:00
Current system time: 2023-11-18 20:43:42-08:00
Hardware address is 5489-9877-2d2a
Last 300 seconds input rate 0 bytes/sec, 0 packets/sec
Last 300 seconds output rate 0 bytes/sec, 0 packets/sec
Input: 1000 bytes, 5 packets
Output: 744 bytes, 3 packets
Input:
  Unicast: 0 packets, Multicast: 0 packets
  Broadcast: 5 packets
Output:
  Unicast: 2 packets, Multicast: 0 packets
  Broadcast: 1 packets
Input bandwidth utilization : 0%
Output bandwidth utilization : 0%
```

command in CLI

features of the interface

statistice on the interface

[Huawei]

Router: interface(2)

3. Commands in interface function view

- Common
 - **display**
 - **undo**
 - ...
- IP function
 - **ip**
 - **ipv6**
 - **nat**
 - **ospf**
 - **rip**
 - ...
- Application Layer function
 - **dhcp**
 - ...
- Testing
 - **ping**
 - **tracert**
 - ...

```
[Huawei-Ethernet0/0/0]dis this
#
interface Ethernet0/0/0
ip address 172.168.1.1 255.255.255.0
dhcp select interface
dhcp server dns-list 114.114.114.114
#
return
[Huawei-Ethernet0/0/0]
```

```
[Huawei-Ethernet0/0/0]ip ?
address          Address
apply            Apply security policy
binding          Enable binding of an interface with a VPN instance
forward-broadcast Specify IP directed broadcast information
netstream        Netstream feature
policy-based-route Enable policy based routing on the interface
relay            DHCP relay
topology          The name of topology
verify           IP verify

[Huawei-Ethernet0/0/0]ip address ?
X.X.X.X          IP address
bootp-alloc      IP address allocated by BOOTP
dhcp-alloc       IP address allocated by DHCP
unnumbered       Share an address with another interface

[Huawei-Ethernet0/0/0]ip address 192.168.1.101 ?
INTEGER<0-32>    Length of IP address mask
X.X.X.X          IP address mask

[Huawei-Ethernet0/0/0]ip address 192.168.1.101
^
Error:Incomplete command found at '^' position.
[Huawei-Ethernet0/0/0]ip address 192.168.1.101 ?
INTEGER<0-32>    Length of IP address mask
X.X.X.X          IP address mask
```


Router: route-forward

The core functions of routers: forward packets according to the routing table

- 1) **search** the **destination IP address** of the packet in the **routing table**
- 2) **find** the corresponding **routing table entry** based on the **longest matching principle**
- 3) **forward packet** according to the routing table entry

```
[Huawei]dis ip routing-table
```

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 6 Routes : 6

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
192.168.1.0/24	Direct	0	0	D	192.168.1.1	Ethernet0/0/0
192.168.1.1/32	Direct	0	0	D	127.0.0.1	Ethernet0/0/0
192.168.2.0/24	Direct	0	0	D	192.168.2.1	Ethernet0/0/1
192.168.2.1/32	Direct	0	0	D	127.0.0.1	Ethernet0/0/1

Router: routing-table(1)

- **Destination/Mask:** Indicates the destination network address and network mask of this route.
- **Protocol:** The protocol type of the route, that is, the protocol through which the router learns about the route.
- **Pre (Preference):** Indicates the routing protocol priority of this route. The highest priority (with the lowest value) will become the current optimal route.
- **Cost(Routing cost):** When multiple routes reaching the same destination have the same routing priority, the one with the lowest routing cost will become the current optimal route.
- **NextHop:** Refers to the next hop address of the destination network that the router points to for this router. This field indicates the next device for data forwarding.
- **Interface:** represents the outbound interface of this route. Indicate which interface of this router the data will be forwarded from.

```
[Huawei]dis ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
      Destinations : 7          Routes : 7

Destination/Mask    Proto   Pre  Cost   Flags NextHop         Interface
-----
127.0.0.0/8         Direct  0    0       D    127.0.0.1         InLoopBack0
127.0.0.1/32        Direct  0    0       D    127.0.0.1         InLoopBack0
192.168.1.0/24       Direct  0    0       D    192.168.1.1       Ethernet0/0/0
192.168.1.1/32       Direct  0    0       D    127.0.0.1         Ethernet0/0/0
192.168.2.0/24       Static  60    0       RD   192.168.3.1       Ethernet0/0/1
192.168.3.0/24       Direct  0    0       D    192.168.3.254     Ethernet0/0/1
192.168.3.254/32    Direct  0    0       D    127.0.0.1         Ethernet0/0/1
```

Router: routing-table(2)

- How to generate the routing table entries in the routing-table ?

- **Direct**

- After the router's interface completes IP address configuration, the corresponding routing table entries are automatically generated

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
192.168.1.0/24	Direct	0	0	D	192.168.1.1	Ethernet0/0/0
192.168.1.1/32	Direct	0	0	D	127.0.0.1	Ethernet0/0/0

- **Static**

192.168.2.0/24	Static	60	0	RD	192.168.3.1	Ethernet0/0/1
----------------	--------	----	---	----	-------------	---------------

- Add through relevant configurations
 - [Huawei] ip route-static 192.168.2.0 255.255.255.0 192.168.3.1
 - » 192.168.2.0 255.255.255.0: Destination and Mask
 - » 192.168.3.1: next hop

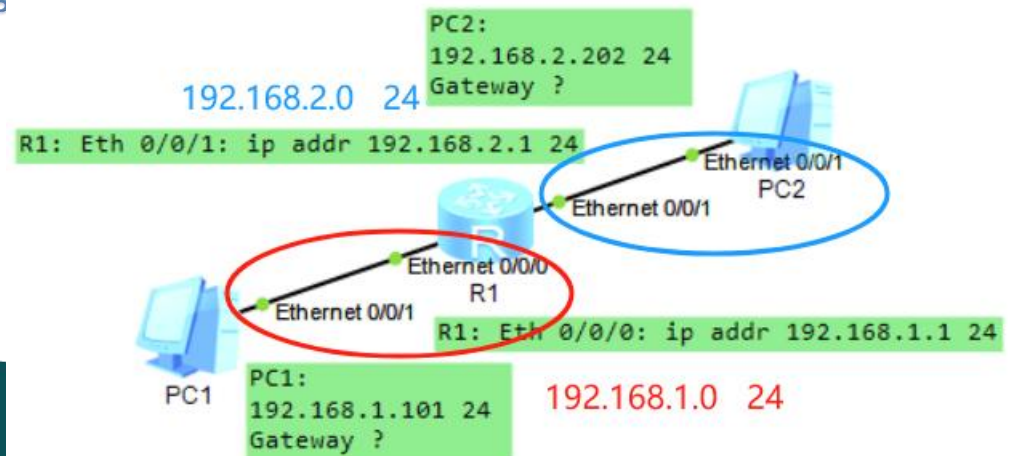
- **Dynamic**

- Generated by routing protocol(RIP, OSPF, BGP...)

Demo1: Direct routing entry(1)

Build a network topology, PC1 and PC2 are not in the same sub-network, do the configuration to make PC1 could send/received packet to/from PC2.

- step1: build the network topology, design the IP settings
 - PC1: 192.168.1.101/24 PC2: 192.168.2.202 /24
 - Ethernet 0/0/0 of R1: in the same sub-network with PC1, 192.168.1.1/24
 - Ethernet 0/0/1 of R2: in the same sub-network with PC1, 192.168.2.1/24
 - How about the configurations of Gateway on PC1 and PC2
 - ?
- step2: Start all the device
- step3: Do the configuration on the PCs and the Router R1
- step4: test by using command “ping”

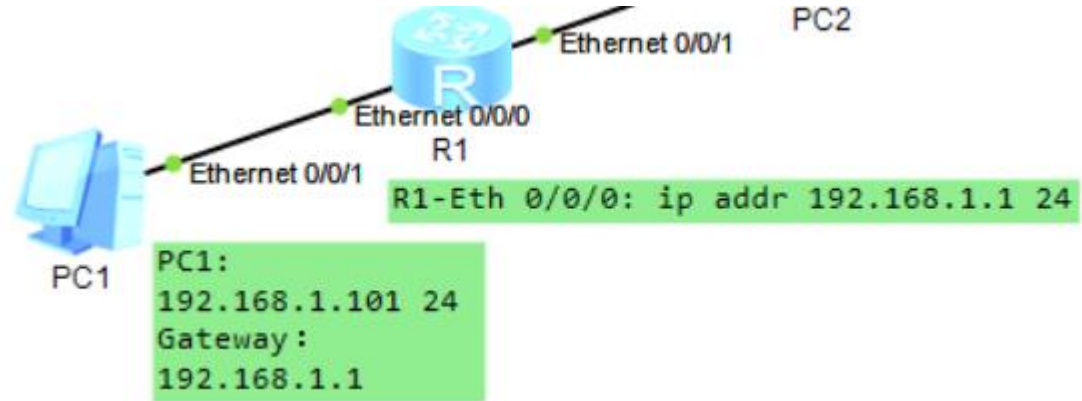


Demo1: Direct routing entry (2)

Analysis:

- PKT1 from **PC1** to **R1-Eth 0/0/0**

1) **PC1** generate **PKT1**(*source IP: 192.168.1.101 , destination IP: 192.168.2.202*)



2) **PC1** caculate the network ID on source and destionation

source IP & subnet mask of source destination IP & subnet mask of source

*souce network ID : 192.168.1.101 & 255.255.255.0 = **192.168.1.0***

*destination network ID : 192.168.2.202 & 255.255.255.0 = **192.168.2.0***

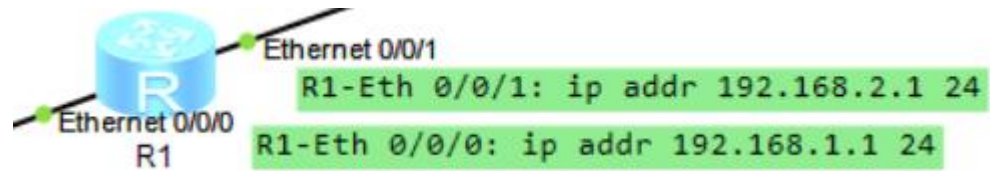
3) **PC1** check if destination network ID equal to souce network ID

✓ *if not, PC1 forward the PKT1 to the Gateway of PC1 (Ethernet 0/0/0 of R1)*

✓ *the Gateway of PC1 could be configured with: 192.168.1.1 (IP address on Ethernet 0/0/0 of R1)*

Demo1: Direct routing entry(3)

- PKT1 from R1-Eth 0/0/0 to R1-Eth 0/0/1



- 1) R1 search the destination IP(192.168.2.202) of PKT1 in the routing-table
- 2) R1 find item "192.168.2.0/24" meet longest matching principle
 - ✓ The matching length with PKT1 destination IP(192.168.2.202) and (192.168.2.0/24) is 24
 - The matching length with PKT1 destination IP(192.168.2.202) and (192.168.2.1/32) is 0
- 3) R1 forward the PKT1 to the NextHop 192.168.2.1 which is on the Eth0/0/1

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 6 Routes : 6

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
192.168.1.0/24	Direct	0	0	D	192.168.1.1	Ethernet0/0/0
192.168.1.1/32	Direct	0	0	D	127.0.0.1	Ethernet0/0/0
192.168.2.0/24	Direct	0	0	D	192.168.2.1	Ethernet0/0/1
192.168.2.1/32	Direct	0	0	D	127.0.0.1	Ethernet0/0/1

Tips:

After the router's **interface** is **configured with an IP address**, the router will directly generate a **direct route** in the routing table based on the correspondence between the address and the interface.

Demo1: Direct routing entry(4)

- PC2 send the PKT2 to PC1

1) PC2 generate PKT2(*source IP: 192.168.2.202 , destination IP: 192.168.1.101*)

2) PC2 caculate the network ID on source and destination

source IP & subnet mask of source destination IP & subnet mask of source

*souce network ID : 192.168.2.202 & 255.255.255.0 = **192.168.2.0***

*destination network ID : 192.168.1.101 & 255.255.255.0 = **192.168.1.0***

3) PC2 check if destination network ID equal to souce network ID

✓ *if not, PC2 forward the PKT2 to the Gateway of PC2 (Ethernet 0/0/1 of R1)*

✓ *the Gateway of PC2 could be configured with: **192.168.2.1** (IP address on Ethernet 0/0/1 of R1)*

4) Route R1 forward the PKT2 to it's interface Ethernet0/0/0 by the PKT2's destinaiton IP address and the routing-table (the related routing entry is generated directly by the Route)



```
[Huawei]dis ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
      Destinations : 6           Routes : 6

Destination/Mask    Proto    Pre  Cost           Flags NextHop         Interface
-----
127.0.0.0/8         Direct   0    0                D    127.0.0.1           InLoopBack0
127.0.0.1/32        Direct   0    0                D    127.0.0.1           InLoopBack0
192.168.1.0/24       Direct   0    0                D    192.168.1.1         Ethernet0/0/0
192.168.1.1/32       Direct   0    0                D    127.0.0.1           Ethernet0/0/0
192.168.2.0/24       Direct   0    0                D    192.168.2.1         Ethernet0/0/1
192.168.2.1/32       Direct   0    0                D    127.0.0.1           Ethernet0/0/1
```

Demo1: Direct routing entry(5)

Q1. If the destination IP of PKT is 192.168.2.1, what would the R1 do after receive the PKT ?

A1. R1 find item “192.168.2.1/32” meet longest matching principle, forward it to the Ethernet0/0/1(destination), 127.0.0.1 is the Inloopback0 interface of R1, R1 receive the PKT, don't forward it out.

The matching length with PKT destination IP(192.168.2.202) and (192.168.2.0/24) is 24

✓ The matching length with PKT destination IP(192.168.2.202) and (192.168.2.1/32) is 32

```
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
      Destinations : 6          Routes : 6

Destination/Mask    Proto    Pre  Cost    Flags NextHop          Interface
-----
127.0.0.0/8        Direct  0     0        D    127.0.0.1        InLoopBack0
127.0.0.1/32       Direct  0     0        D    127.0.0.1        InLoopBack0
192.168.1.0/24     Direct  0     0        D    192.168.1.1      Ethernet0/0/0
192.168.1.1/32     Direct  0     0        D    127.0.0.1        Ethernet0/0/0
192.168.2.0/24     Direct  0     0        D    192.168.2.1      Ethernet0/0/1
192.168.2.1/32     Direct  0     0        D    127.0.0.1        Ethernet0/0/1
```

Q2. If the destination IP of PKT couldn't match the items of the routing-table, what would the R1 do after receive the PKT ?

A2. R1 drop the PKT

Q3. Does the router know the routing information to all destinations? Is there any way to forward packets without knowing it's rout?

Demo1: Direct routing entry(6)

Q3. Does the router know the routing information to all destinations? Is there any way to forward packets without knowing it's route?

A3. some tips from the routing-table on the end device(e.t. laptop)

Default route is the route selected by the router when no other route exists for the destination address in the IP packet. All packets whose destination is not in the router's routing table will use the default route.

here two items with **0.0.0.0** as **target** and **0.0.0.0** as **mask** is the **default route**

命令提示符

无线局域网适配器 WLAN:

```
连接特定的 DNS 后缀 . . . . . :  
本地链接 IPv6 地址. . . . . : fe80::...%18  
IPv4 地址 . . . . . : 192.168.100.31  
子网掩码 . . . . . : 255.255.255.0  
默认网关. . . . . : 192.168.100.1
```

移动宽带适配器 手机网络: Mobile broadband adapter mobile network

```
连接特定的 DNS 后缀 . . . . . :  
IPv6 地址 . . . . . : ::e4...  
临时 IPv6 地址. . . . . : ::2...  
本地链接 IPv6 地址. . . . . : fe80::8...%16  
IPv4 地址 . . . . . : 10.123.182.11  
子网掩码 . . . . . : 255.255.255.0  
默认网关. . . . . : fe80::...%16  
Gateway 10.123.182.1
```

C:\Users\sustech>route PRINT -4

接口列表

```
15... Intel(R) Ethernet Connection (10) I219-V  
25... VirtualBox Host-Only Ethernet Adapter  
32... Microsoft Wi-Fi Direct Virtual Adapter #2  
11... Microsoft Wi-Fi Direct Virtual Adapter #3  
18... Intel(R) Wireless-AC 9560 160MHz  
16... Generic Mobile Broadband Adapter  
20... Bluetooth Device (Personal Area Network)  
1... Software Loopback Interface 1
```

IPv4 路由表

IPv4 Routing-table on Laptop

活动路由:	target	mask	Gateway	Interface	
网络目标	网络掩码	网关	接口	跃点数	
	0.0.0.0	0.0.0.0	10.123.182.1	10.123.182.11	311
	0.0.0.0	0.0.0.0	192.168.100.1	192.168.100.31	50

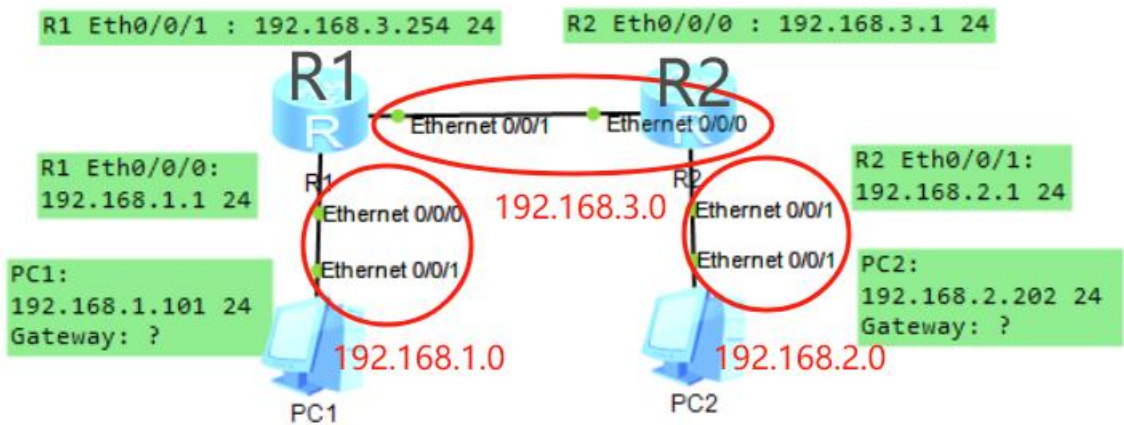
Demo2: static routing entry(1)

- Build a network topology, there are 2 PCs(PC1 and PC2) and 2 Routers(R1 and R2), the connects and the basic configuration are as shown in the following figure, complete the configurations to make PC1 could send/ receive PKT to/from PC2.

Q1. What's the configuration on Gateway of PC1 and PC2?

Q2. What's the direct routing entries of R1 and R2?

Q3. While R1 receives the PKT(it's destination is PC2) from PC1, could R1 forward the PKT to PC2 directly? would the direct routing entry in R1 help it to do it?

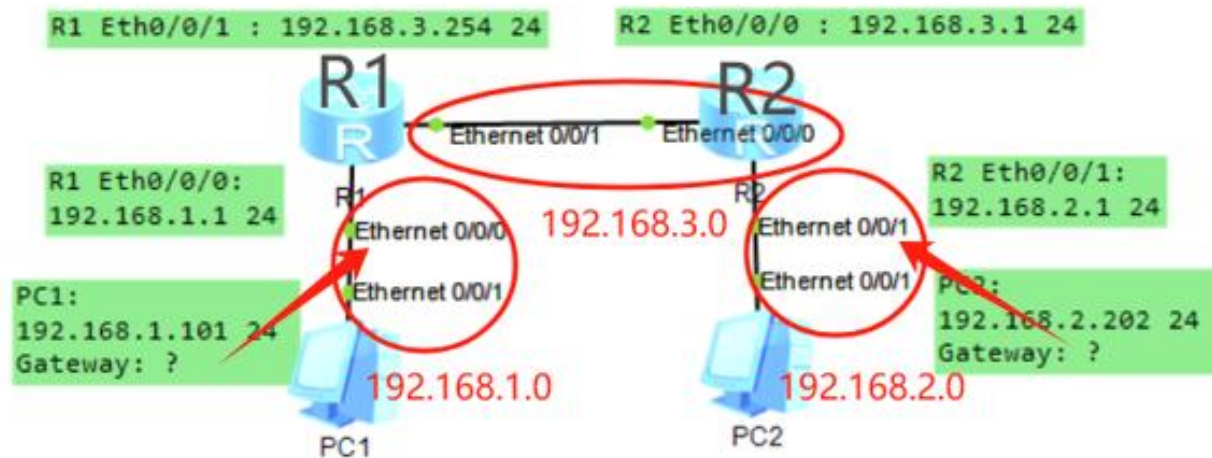


Q4. While R2 receives the PKT(it's destination is PC1) from PC2, could R2 forward the PKT to PC1 directly? would the direct routing entry in R2 help it to do it?

Demo2: static routing entry(2)

- Build a network topology, there are 2 PCs(PC1 and PC2) and 2 Routers(R1 and R2), the connects and the basic configuration are as shown in the following figure, complete the configurations to make PC1 could send/ receive PKT to/from PC2.

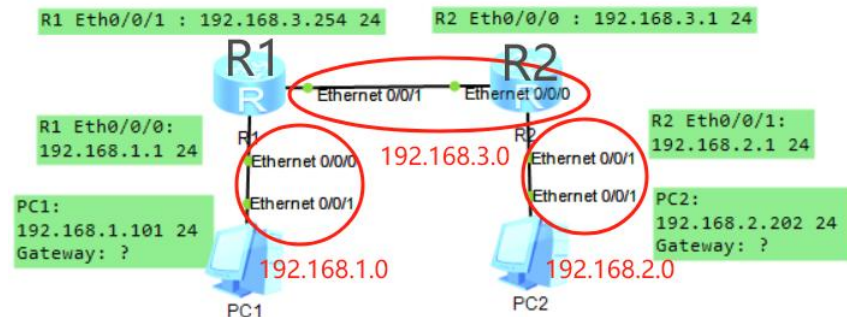
Q1. What's the configuration on Gateway of PC1 and PC2?



Demo2: static routing entry(3)

- Build a network topology, there are 2 PCs(PC1 and PC2) and 2 Routers(R1 and R2), the connects and the basic configuration are as shown in the following figure, complete the configurations to make PC1 could send/ receive PKT to/from PC2.

Q2. What's the direct routing entries of R1 and R2?



R1						
[Huawei]dis ip routing-table						
Route Flags: R - relay, D - download to fib						

Routing Tables: Public						
Destinations : 6			Routes : 6			
Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
192.168.1.0/24	Direct	0	0	D	192.168.1.1	Ethernet0/0/0
192.168.1.1/32	Direct	0	0	D	127.0.0.1	Ethernet0/0/0
192.168.3.0/24	Direct	0	0	D	192.168.3.254	Ethernet0/0/1
192.168.3.254/32	Direct	0	0	D	127.0.0.1	Ethernet0/0/1

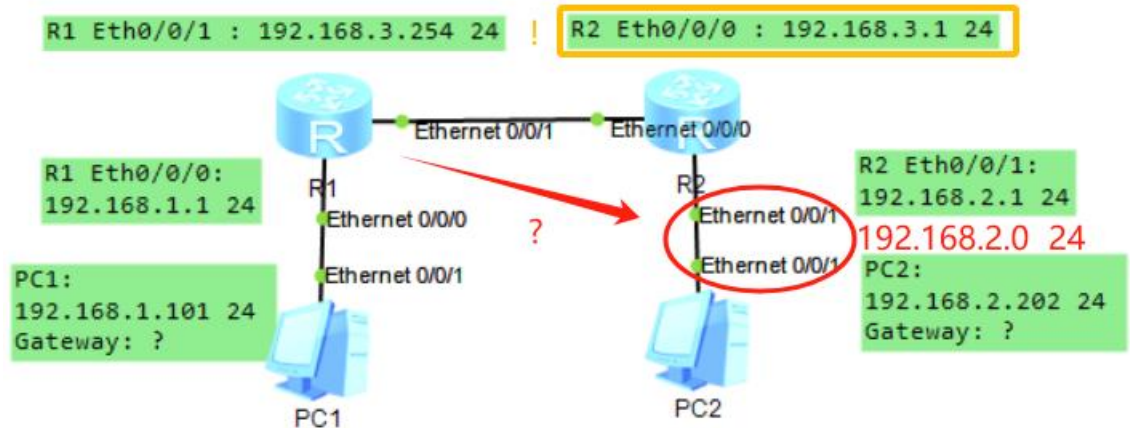
R2						
[Huawei]dis ip routing-table						
Route Flags: R - relay, D - download to fib						

Routing Tables: Public						
Destinations : 6			Routes : 6			
Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
192.168.2.0/24	Direct	0	0	D	192.168.2.1	Ethernet0/0/1
192.168.2.1/32	Direct	0	0	D	127.0.0.1	Ethernet0/0/1
192.168.3.0/24	Direct	0	0	D	192.168.3.1	Ethernet0/0/0
192.168.3.1/32	Direct	0	0	D	127.0.0.1	Ethernet0/0/0

Demo2: static routing entry(4)

Q3. While R1 receives the PKT(its destination is PC2) from PC1, could R1 forward the PKT to PC2 directly? would the direct routing entry in R1 help it to do it?

A3. No, No



Q4. While R2 receives the PKT(its destination is PC1) from PC2, could R2 forward the PKT to PC1 directly? would the direct routing entry in R2 help it to do it?

A4. Same as R1

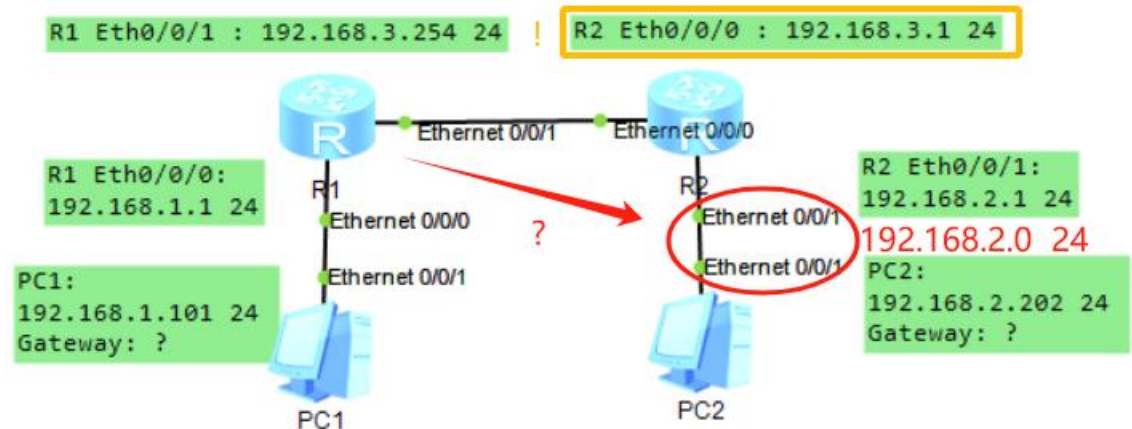
R1						
[Huawei]dis ip routing-table						
Route Flags: R - relay, D - download to fib						

Routing Tables: Public						
Destinations : 6			Routes : 6			
Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
192.168.1.0/24	Direct	0	0	D	192.168.1.1	Ethernet0/0/0
192.168.1.1/32	Direct	0	0	D	127.0.0.1	Ethernet0/0/0
192.168.3.0/24	Direct	0	0	D	192.168.3.254	Ethernet0/0/1
192.168.3.254/32	Direct	0	0	D	127.0.0.1	Ethernet0/0/1

Demo2: static routing entry(5)

In this case, static routing entry could help the Route the forward the PKT to the next hop.

- **R1** forward the PKT (**destination network ID is 192.168.2.0**, mask is **255.255.255.0**) to the interface **Eth0/0/0(192.168.3.1)** of **R2**
 - the IP address(**192.168.3.1**) of interface **Eth0/0/0** on **R2** is the **next hop**
 - the interface **Eth0/0/0(192.168.3.1)** of **R2** is in the same subnet as the **interface Eth0/0/1 of R1**, it could be reached directly without other router's forwarding
- After the PKT reaches **R2**, **R2** send the PKT according to its direct routing entry.



```
[Huawei]ip route-static 192.168.2.0 24 192.168.3.1
```

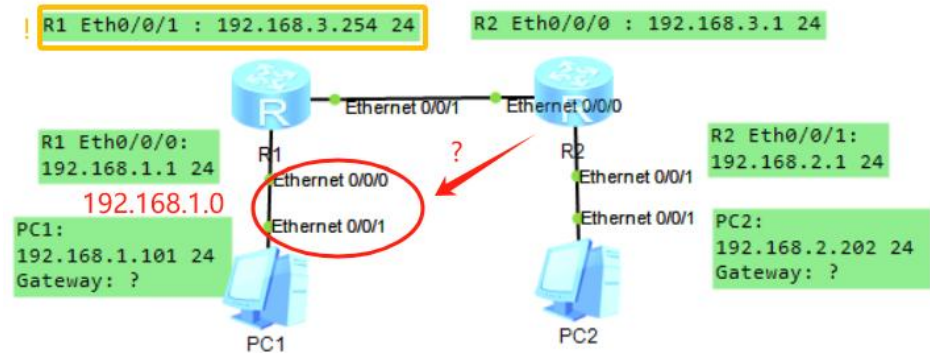
R1							
<Huawei>dis ip routing-table							
Route Flags: R - relay, D - download to fib							

Routing Tables: Public							
Destinations : 7				Routes : 7			
Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface	
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0	
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0	
192.168.1.0/24	Direct	0	0	D	192.168.1.1	Ethernet0/0/0	
192.168.1.1/32	Direct	0	0	D	127.0.0.1	Ethernet0/0/0	
192.168.2.0/24	Static	60	0	RD	192.168.3.1	Ethernet0/0/1	
192.168.3.0/24	Direct	0	0	D	192.168.3.254	Ethernet0/0/1	
192.168.3.254/32	Direct	0	0	D	127.0.0.1	Ethernet0/0/1	

Router: static routing entry(6)

In this case, static routing entry could help the router to forward the PKT to the next hop.

- **R2** forward the PKT (**destination network ID is 192.168.1.0, mask is 255.255.255.0**) to the interface Eth0/0/1(192.168.3.254) of R1
 - the IP address(**192.168.3.254**) of interface Eth0/0/1 on R1 is the **next hop**
 - the interface Eth0/0/1(192.168.3.254) of R1 is in the same subnet as the **interface Eth0/0/0 of R2**, it could be reached directly without other router's forwarding
- After the PKT reaches R1, R1 send the PKT according to its direct routing entry.



```
[Huawei]ip route-static 192.168.1.0 24 192.168.3.254
```

R2						
[Huawei]dis ip routing-table						
Route Flags: R - relay, D - download to fib						

Routing Tables: Public						
Destinations : 7			Routes : 7			
Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
192.168.1.0/24	Static	60	0	RD	192.168.3.254	Ethernet0/0/0
192.168.2.0/24	Direct	0	0	D	192.168.2.1	Ethernet0/0/1
192.168.2.1/32	Direct	0	0	D	127.0.0.1	Ethernet0/0/1
192.168.3.0/24	Direct	0	0	D	192.168.3.1	Ethernet0/0/0
192.168.3.1/32	Direct	0	0	D	127.0.0.1	Ethernet0/0/0
[Huawei]						

Practice1. static routing entry

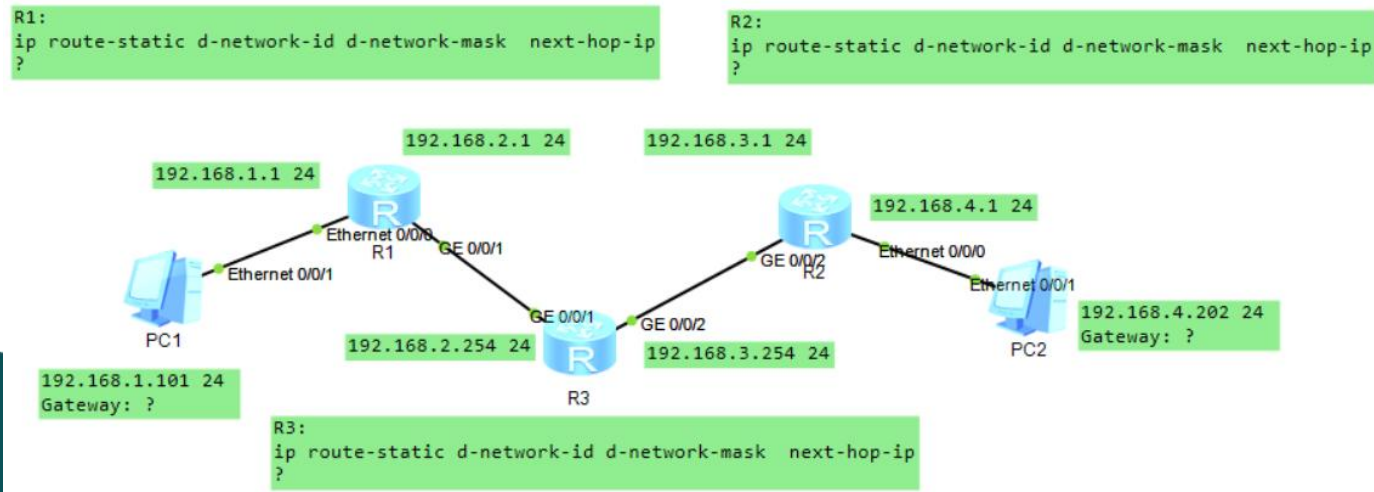
Build a network topology, there are 2 PCs(PC1 and PC2) and 3 Routers(R1, R2 and R3), the connects and the basic configuration are as shown in the following figure, complete the configurations to make PC1 could send/ receive PKT to/from PC2.

– NOTE:

- Show all interface in the topology
- the configuration info are strongly suggested to record in the “text” of the topology



- Show the configuration on PCs
- Show the ip routing-table on each Route
 - Describe the specific role of routing table entries in PKT forwarding
- Testing by command “ping”



Router: DHCP service

Router could provide DHCP service, it work as DHCP server or relay the DHCP related packet.

- DHCP is built on a Client-Server model
 - server: a host providing initialization parameters through DHCP
 - client: a host requesting initialization parameters from a DHCP server
 - designated DHCP server hosts allocate network addresses and deliver configuration parameters to dynamically configured hosts
- BOOTP is a transport mechanism for a collection of configuration information. BOOTP using port 67 AND 68 of UDP.

```
C:\Windows\system32\cmd.exe
无线局域网适配器 WLAN:

连接特定的 DNS 后缀 . . . . . :
描述. . . . . : Intel(R) Dual Band Wireless-AC 8265
物理地址. . . . . : 88-81-4E-58-80-50
DHCP 已启用 . . . . . : 是
自动配置已启用. . . . . : 是
本地连接 IPv6 地址. . . . . : fe80::84bf:7fhe:b61f:c23b%19(首选)
IPv4 地址. . . . . : 192.168.2.104(首选)
子网掩码. . . . . : 255.255.255.0
获得租约的时间. . . . . : 2020年11月6日 18:44:01
租约过期的时间. . . . . : 2020年11月7日 18:44:01
默认网关. . . . . : 192.168.2.1
DHCP 服务器. . . . . : 192.168.2.1
DHCPv6 IAID. . . . . : 277897646
DHCPv6 客户端 DUID. . . . . : 00-01-00-01-00-00-00-00-00-00-00-00-00-00-00-00
DNS 服务器. . . . . : 116.77.76.254
                        116.77.76.253
```

default gateway, DHCP Server

DHCP

RFC 2131 Dynamic Host Configuration Protocol March 1997

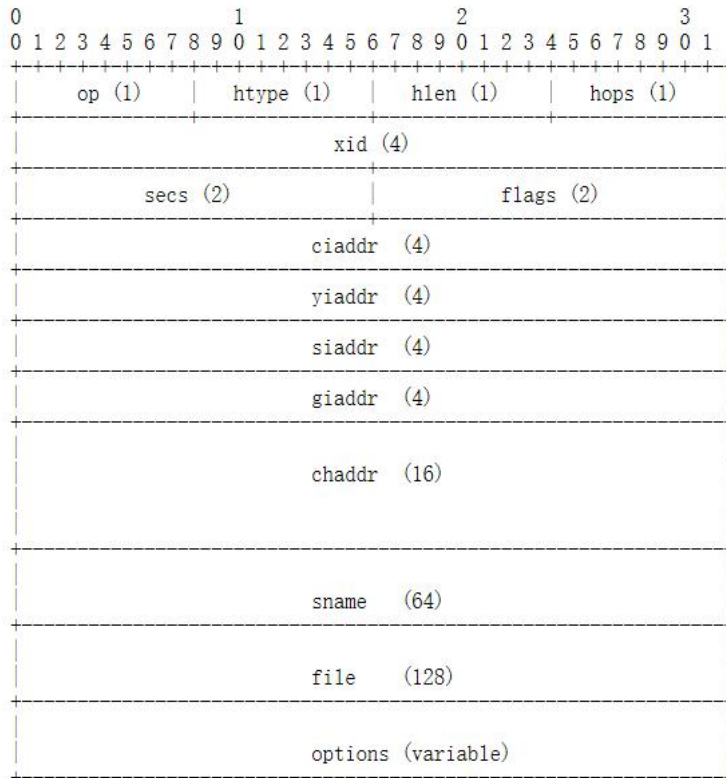


Figure 1: Format of a DHCP message

FIELD	OCTETS	DESCRIPTION
op	1	Message op code / message type. 1 = BOOTREQUEST, 2 = BOOTREPLY
htype	1	Hardware address type, see ARP section in "Assigned Numbers" RFC; e.g., '1' = 10mb ethernet.
hlen	1	Hardware address length (e.g. '6' for 10mb ethernet).
hops	1	Client sets to zero, optionally used by relay agents when booting via a relay agent.
xid	4	Transaction ID, a random number chosen by the client, used by the client and server to associate messages and responses between a client and a server.
secs	2	Filled in by client, seconds elapsed since client began address acquisition or renewal process.
flags	2	Flags (see figure 2).
ciaddr	4	Client IP address; only filled in if client is in BOUND, RENEW or REBINDING state and can respond to ARP requests.
yiaddr	4	'your' (client) IP address.
siaddr	4	IP address of next server to use in bootstrap; returned in DHCPDISCOVER, DHCPOFFER by server.
giaddr	4	Relay agent IP address, used in booting via a relay agent.
chaddr	16	Client hardware address.
sname	64	Optional server host name, null terminated string.
file	128	Boot file name, null terminated string; "generic" name or null in DHCPDISCOVER, fully qualified directory-path name in DHCPOFFER.
options	var	Optional parameters field. See the options documents for a list of defined options.

Table 1: Description of fields in a DHCP message

DHCP Session(1)

- Client-Server interaction when allocating a new network address

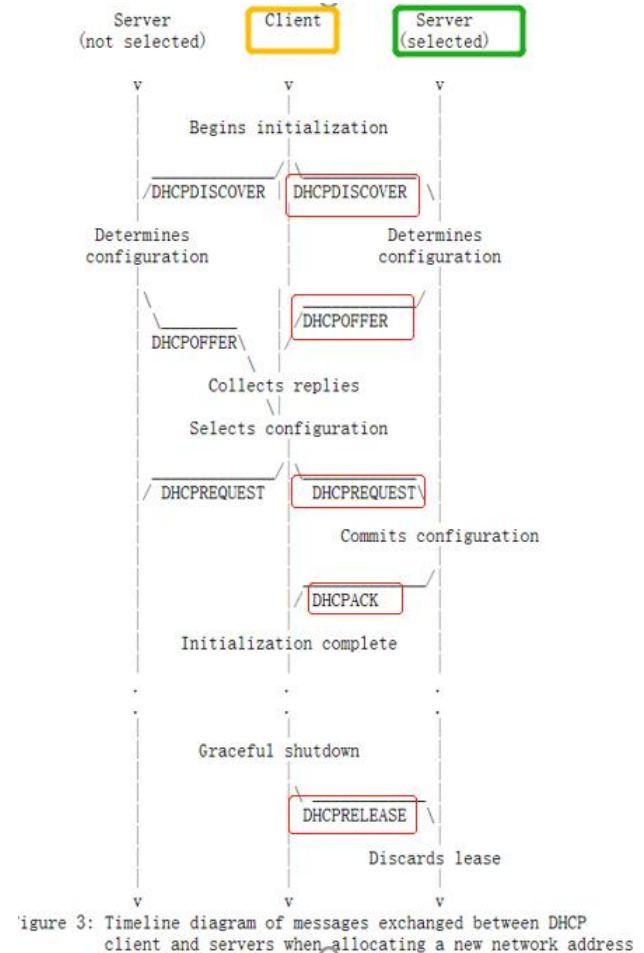
Source	Destination	Protocol	Info
0.0.0.0	255.255.255.255	DHCP	DHCP Discover - Transaction ID 0x3e5e0ce3
192.168.1.1	255.255.255.255	DHCP	DHCP Offer - Transaction ID 0x3e5e0ce3
0.0.0.0	255.255.255.255	DHCP	DHCP Request - Transaction ID 0x3e5e0ce3
192.168.1.1	255.255.255.255	DHCP	DHCP ACK - Transaction ID 0x3e5e0ce3
192.168.1.101	192.168.1.1	DHCP	DHCP Request - Transaction ID 0x257e55a3
192.168.1.1	255.255.255.255	DHCP	DHCP ACK - Transaction ID 0x257e55a3
192.168.1.101	192.168.1.1	DHCP	DHCP Release - Transaction ID 0xb7a32733

Tips in command line:

*While network interface card is set as DHCP client,
using 'ipconfig /renew' to request a dynamically assigned IP addresses.
using 'ipconfig /release' to release the dynamically assigned IP addresses.*

Tips in Wireshark display filter : DHCP or

udp.port == 67 || udp.port == 68



DHCP Discover

```
> Frame 2: 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits)
> Ethernet II, Src: Dell_4f:36:23 (00:08:74:4f:36:23), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
> Internet Protocol Version 4, Src: 0.0.0.0 (0.0.0.0), Dst: 255.255.255.255 (255.255.255.255)
> User Datagram Protocol, Src Port: bootpc (68), Dst Port: bootps (67)
v Bootstrap Protocol (Discover)
```

```
    Message type: Boot Request (1)
    Hardware type: Ethernet (0x01)
    Hardware address length: 6
    Hops: 0
    Transaction ID: 0x3e5e0ce3
    Seconds elapsed: 0
> Bootp flags: 0x0000 (Unicast)
    Client IP address: 0.0.0.0 (0.0.0.0)
    Your (client) IP address: 0.0.0.0 (0.0.0.0)
    Next server IP address: 0.0.0.0 (0.0.0.0)
    Relay agent IP address: 0.0.0.0 (0.0.0.0)
    Client MAC address: Dell_4f:36:23 (00:08:74:4f:36:23)
    Client hardware address padding: 00000000000000000000
    Server host name not given
    Boot file name not given
    Magic cookie: DHCP
> Option: (53) DHCP Message Type (Discover)
> Option: (116) DHCP Auto-Configuration
> Option: (61) Client identifier
> Option: (50) Requested IP Address
> Option: (12) Host Name
> Option: (60) Vendor class identifier
> Option: (55) Parameter Request List
> Option: (255) End
    Padding: 00000000000000000000
```

```
v Option: (53) DHCP Message Type (Discover)
    Length: 1
    DHCP: Discover (1)
v Option: (116) DHCP Auto-Configuration
    Length: 1
    DHCP Auto-Configuration: AutoConfigure (1)
v Option: (61) Client identifier
    Length: 7
    Hardware type: Ethernet (0x01)
    Client MAC address: Dell_4f:36:23 (00:08:74:4f:36:23)
v Option: (50) Requested IP Address
    Length: 4
    Requested IP Address: 192.168.1.101 (192.168.1.101)
v Option: (12) Host Name
    Length: 4
    Host Name: Noho
v Option: (60) Vendor class identifier
    Length: 8
    Vendor class identifier: MSFT 5.0
v Option: (55) Parameter Request List
    Length: 11
    Parameter Request List Item: (1) Subnet Mask
    Parameter Request List Item: (15) Domain Name
    Parameter Request List Item: (3) Router
    Parameter Request List Item: (6) Domain Name Server
    Parameter Request List Item: (44) NetBIOS over TCP/IP Name Server
    Parameter Request List Item: (46) NetBIOS over TCP/IP Node Type
    Parameter Request List Item: (47) NetBIOS over TCP/IP Scope
    Parameter Request List Item: (31) Perform Router Discover
    Parameter Request List Item: (33) Static Route
    Parameter Request List Item: (249) Private/Classless Static Route (Microsoft)
```

DHCP Offer

```
> User Datagram Protocol, Src Port: bootps (67), Dst Port: bootpc (68)
```

- ✓ Bootstrap Protocol (Offer)

Message type: Boot Reply (2)

Hardware type: Ethernet (0x01)

Hardware address length: 6

Hops: 0

Transaction ID: 0x3e5e0ce3

Seconds elapsed: 0

```
> Bootp flags: 0x0000 (Unicast)
```

Client IP address: 0.0.0.0 (0.0.0.0)

```
Your (client) IP address: 192.168.1.101 (192.168.1.101)
```

Next server IP address: 0.0.0.0 (0.0.0.0)

Relay agent IP address: 0.0.0.0 (0.0.0.0)

Client MAC address: Dell 4f:36:23 (00:08:74:4f:36:23)

```
Client hardware address padding: 00000000000000000000
```

Server host name not given

Boot file name not given

Magic cookie: DHCP

✓ Option: (53) DHCP Message Type (Offer)

Length: 1

DHCP: Offer (2)

- Option: (1) Subnet Mask

Length: 4

Subnet Mask: 255.255.255.0

- Option: (3) Router

Length: 4

Router: 192.168.1.1 (192.168.1.1)

- Option: (6) Domain Name Server

Length: 8

Domain Name Server: ns10.attbi.com (63.240.76.19)

Domain Name Server: 204.127.198.19 (204.127.198.19)

✓ Option: (15) Domain Name

Length: 22

Domain Name: ne2.client2.attbi.com

- Option: (51) IP Address Lease Time

Length: 4

IP Address Lease Time: (86400s) 1 day

- Option: (54) DHCP Server Identifier

Length: 4

DHCP Server Identifier: 192.168.1.1 (192.168.1.1)

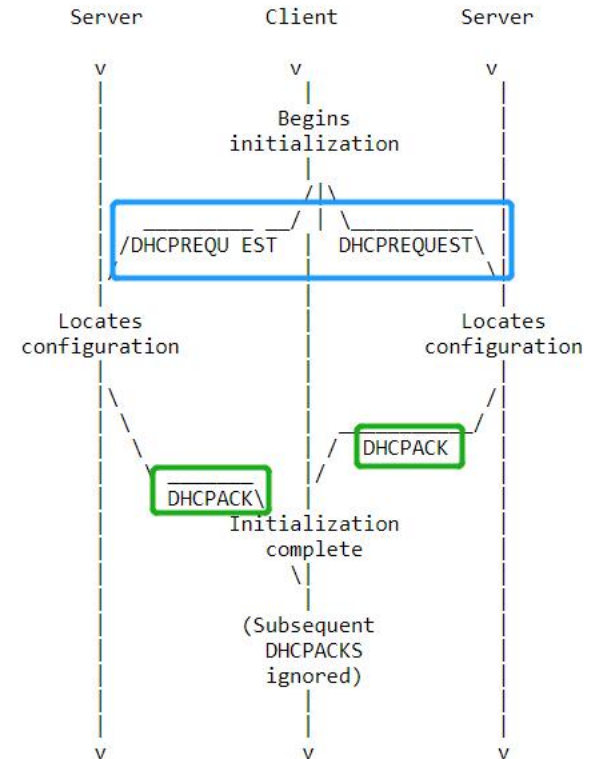
✓ Option: (255) End

Option End: 255

[illegible]

DHCP Session(2)

- Client-Server interaction when reusing a previously allocated network address



dhcp			
Source	Destination	Protocol	Info
activate.adobe.com	255.255.255.255	DHCP	DHCP Request - Transaction ID 0x98bd1be8
192.168.2.1	LAPTOP-RITC8FUU.local	DHCP	DHCP ACK - Transaction ID 0x98bd1be8

DHCP Request & Ack

```
> User Datagram Protocol, Src Port: bootpc (68), Dst Port: bootps (67)
v Dynamic Host Configuration Protocol (Request)
  Message type: Boot Request (1)
  Hardware type: Ethernet (0x01)
  Hardware address length: 6
  Hops: 0
  Transaction ID: 0x98bd1be8
  Seconds elapsed: 0
  > Bootp flags: 0x0000 (Unicast)
  Client IP address: activate.adobe.com (0.0.0.0)
  Your (client) IP address: activate.adobe.com (0.0.0.0)
  Next server IP address: activate.adobe.com (0.0.0.0)
  Relay agent IP address: activate.adobe.com (0.0.0.0)
  Client MAC address: LAPTOP-RITC8FUU.local (90:61:ae:5c:69:58)
  Client hardware address padding: 00000000000000000000
  Server host name not given
  Boot file name not given
  Magic cookie: DHCP
  > Option: (53) DHCP Message Type (Request)
  > Option: (61) Client identifier
  > Option: (50) Requested IP Address (192.168.2.104)
  > Option: (12) Host Name
  > Option: (81) Client Fully Qualified Domain Name
  > Option: (60) Vendor class identifier
  > Option: (55) Parameter Request List
  > Option: (255) End
```

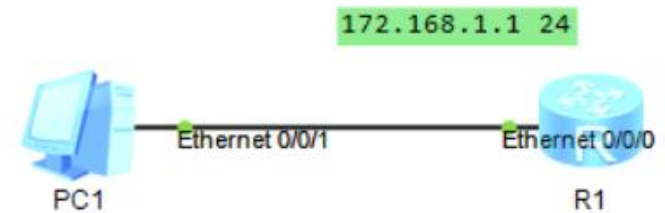
```
User Datagram Protocol, Src Port: bootps (67), Dst Port: bootpc (68)
Dynamic Host Configuration Protocol (ACK)
  Message type: Boot Reply (2)
  Hardware type: Ethernet (0x01)
  Hardware address length: 6
  Hops: 0
  Transaction ID: 0x98bd1be8
  Seconds elapsed: 0
  > Bootp flags: 0x0000 (Unicast)
  Client IP address: activate.adobe.com (0.0.0.0)
  Your (client) IP address: LAPTOP-RITC8FUU.local (192.168.2.104)
  Next server IP address: 192.168.2.1 (192.168.2.1)
  Relay agent IP address: activate.adobe.com (0.0.0.0)
  Client MAC address: LAPTOP-RITC8FUU.local (90:61:ae:5c:69:58)
  Client hardware address padding: 00000000000000000000
  Server host name not given
  Boot file name not given
  Magic cookie: DHCP
  > Option: (53) DHCP Message Type (ACK)
  > Option: (1) Subnet Mask (255.255.255.0)
  > Option: (2) Time Offset
  > Option: (3) Router
  > Option: (23) Default IP Time-to-Live
  > Option: (51) IP Address Lease Time
  > Option: (54) DHCP Server Identifier (192.168.2.1)
  > Option: (6) Domain Name Server
  > Option: (58) Renewal Time Value
  > Option: (59) Rebinding Time Value
  > Option: (255) End
  Padding: 00
```


Router: DHCP service(1-1)

- How to make a Router provide the DHCP service ?
 - **method1: Provide DHCP services on the interface**
 - method2: Create a global address pool for routers to provide DHCP services

method1(in the Router)

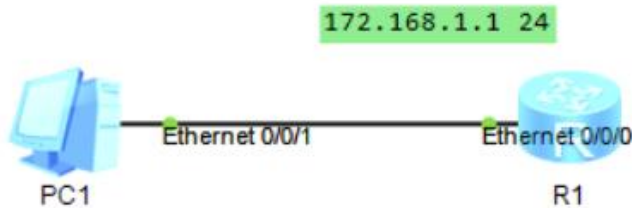
- make DHCP enable [Huawei]
 - command: **dhcp enable**
- select the interface
 - the interface is most near the DHCP client, here is "Ethernet 0/0/1"
 - the interface of the router works as the Gateway of the DHCP client
- Configure the interface, here is [Huawei-Ethernet0/0/0]
 - IP address
 - command : **ip address 172.168.1.1 255.255.255.0**
 - IP address of the interface would be the gateway
 - to determine the DHCP service is provided by the interface
 - command : **dhcp select interface**
 - to point out the DNS-list
 - command : **dhcp server dns-list 114.114.114.114**
 - lease time ...
 - Restricted address range ...



tips: Don't forget to set the PC as DHCP client

Router: DHCP service(1-2)

- Testing
 - by using command “ipconfig” on PC



PC1 configuration window showing the DHCP settings. The DHCP option is selected under IPv4 Configuration. The DNS server address is set to 114.114.114.114. The Apply button is highlighted with a red arrow.

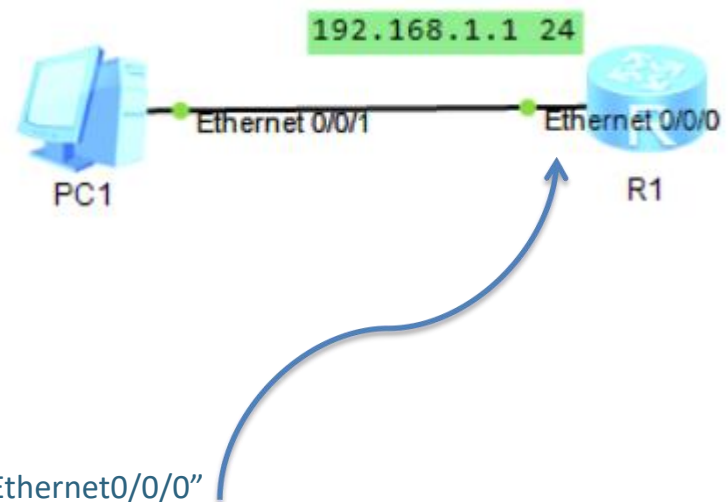
```
PC1
Basic Config  Command  MCPacket  UdpPacket  Console
Welcome to use PC Simulator!

PC>ipconfig

Link local IPv6 address.....: fe80::5689:98ff:fe8c:61eb
IPv6 address.....: :: / 128
IPv6 gateway.....: ::
IPv4 address.....: 172.168.1.254
Subnet mask.....: 255.255.255.0
Gateway.....: 172.168.1.1
Physical address.....: 54-89-98-8C-61-EB
DNS server.....: 114.114.114.114
```

Router: DHCP service(2-1)

- How to make a Route provide the DHCP service?
 - **method2:Create a global address pool for routers to use to provide DHCP services**
 - make DHCP enable [Huawei]
 - command: **dhcp enable**
 - make a “ip pool” [Huawei]
 - command: **ip pool [ip-pool-name]**
 - point out network
 - command: **network 192.168.1.0**
 - point out Gateway and Mask
 - command: **gateway-list 192.168.1.1**
 - point out the DNS-list
 - command : **dns-list 114.114.114.114**
 - point out lease time and restricted address range ...
 - select the interface
 - the interface is most near the DHCP client, here is “Ethernet0/0/0”
 - Configure the interface, here is [Huawei-Ethernet0/0/0]
 - IP address
 - command : **ip address 192.168.1.1 255.255.255.0**
 - to determine the DHCP service is provide by the ip pool
 - command : **dhcp select global**



Router: DHCP service(2-2)

```
[Huawei]dis ip pool
```

```
-----  
Pool-name      : test  
Pool-No       : 0  
Position      : Local          Status      : Unlocked  
Gateway-0     : 10.1.0.1  
Mask          : -  
VPN instance  : --
```

IP address Statistic

```
Total      :0  
Used       :0      Idle      :0  
Expired    :0      Conflict  :0      Disable   :0
```

```
[Huawei]
```

```
[Huawei-ip-pool-test]dis this
```

```
#  
ip pool test  
gateway-list 192.168.1.1  
network 192.168.1.0 mask 255.255.255.0  
excluded-ip-address 192.168.1.200  
lease day 1 hour 12 minute 0  
dns-list 114.114.114.114
```

```
#
```

```
return
```

```
[Huawei-ip-pool-test]
```

```
[Huawei-Ethernet0/0/0]dis this
```

```
#  
interface Ethernet0/0/0  
ip address 10.1.0.1 255.255.0.0  
dhcp select global
```

```
#
```

```
return
```

```
[Huawei-Ethernet0/0/0]
```

```
PC>ipconfig /renew
```

IP Configuration

```
Link local IPv6 address.....: fe80::5689:98ff:fee8:4d85  
IPv6 address.....: :: / 128  
IPv6 gateway.....: ::  
IPv4 address.....: 192.168.1.254  
Subnet mask.....: 255.255.255.0  
Gateway.....: 192.168.1.1  
Physical address.....: 54-89-98-E8-4D-85  
DNS server.....: 114.114.114.114
```

```
PC>ping 192.168.1.1
```

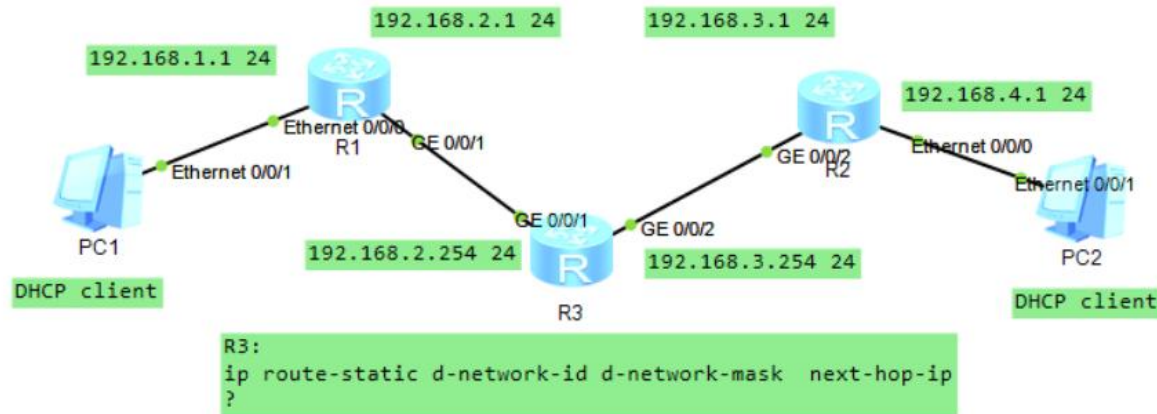
```
Ping 192.168.1.1: 32 data bytes, Press Ctrl_C to break  
From 192.168.1.1: bytes=32 seq=1 ttl=255 time=32 ms  
192.168.1.1: bytes=32 seq=2 ttl=255 time=31 ms
```


Practic2: DHCP server on Router

- Set up the DHCP service on the Router in the existing network topology(practice 1), set the PC as the DHCP client
- test whether the PC has obtained relevant configurations, and test network connectivity

```
R1:  
ip route-static d-network-id d-network-mask next-hop-ip  
?
```

```
R2:  
ip route-static d-network-id d-network-mask next-hop-ip  
?
```



Part B: Cloud in eNSP



“Cloud”

- ✓ “Cloud” in eNSP could communicate with virtual or real network cards on the PC where eNSP is located.
- ✓ With the help of the “Cloud” and network card, virtual network in eNSP could communicate with the network card which works on the PC where eNSP is running.

Prerequisite: 1) eNSP is installed on the PC; 2) there is an idle network cards on the PC;

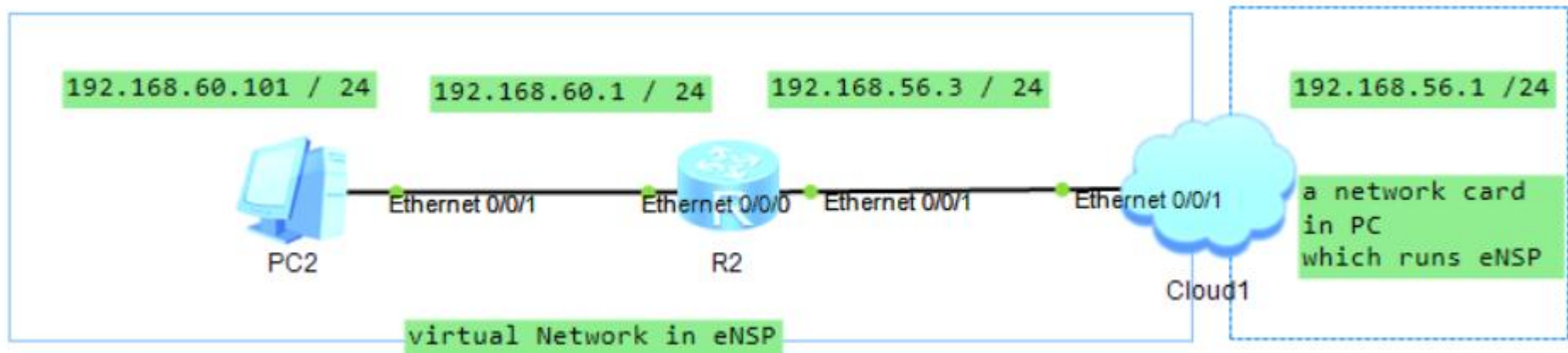
Specific steps:

- 1) **Choose** the idle network card(.e.g. “eth5”) on the PC, find its configuration(IP address, Gateway .etc.)
- 2) **Build** a network topology by eNSP(.e.g. named as “test.topo”) .
- 3) **Add** a “Cloud” device to the topology, **Configure** the “Cloud” device in the “test.topo” of eNSP with “Port Mapping”.
- 4) **Connect** the “Cloud” device with the network in the topology.

Interconnection(Demo)

Demo: virtual network in eNSP could communicate with the network card which works on the PC where eNSP is running :

- 1) **Choose** the idle network card.
- 2) **Build** a network topology by eNSP, consisting of a “Cloud” device, one PC(PC2), and a Router(R2). **Configure the devices.**
- 3) **Add** a “Cloud” device(Cloud1) to the topology, **Configure** the “Cloud1” .
- 4) **Connect** the “Cloud” device(Cloud1) with the network in the topology.
- 5) **Testing:** invok “ping” test on PC2 to 192.168.56.1, or invoke “ping” test on 192.168.56.1 to PC2.



Interconnection(Demo-step1)

1) **Choose** the idle network card.

In this demo, the IPv4 configuration of the networkcard is:

IP address :192.168.56.1

Subnetwork: 255.255.255.0

Gateway: 192.168.56.3

```
C:\Users\sustech>ipconfig
```

```
Windows IP 配置
```

```
以太网适配器 以太网 2:
```

```
媒体状态 . . . . . : 媒体已断开连接
连接特定的 DNS 后缀 . . . . . :
```

```
以太网适配器 VirtualBox Host-Only Network:
```

```
连接特定的 DNS 后缀 . . . . . :
本地链接 IPv6 地址. . . . . : fe80::bc4d:7442:ab2:3230%29
自动配置 IPv4 地址 . . . . . : 169.254.120.101
子网掩码 . . . . . : 255.255.0.0
默认网关. . . . . :
```

```
以太网适配器 以太网 4:
```

```
连接特定的 DNS 后缀 . . . . . :
本地链接 IPv6 地址. . . . . : fe80::a26b:3cb4:6c5b:d498%25
IPv4 地址 . . . . . : 13.0.0.101
子网掩码 . . . . . : 255.0.0.0
默认网关. . . . . : 13.0.0.254
```

```
以太网适配器 以太网 5:
```

```
连接特定的 DNS 后缀 . . . . . :
IPv6 地址 . . . . . : 2025::101
本地链接 IPv6 地址. . . . . : fe80::82b3:f45e:5f37:cad0%10
IPv4 地址 . . . . . : 192.168.56.1
子网掩码 . . . . . : 255.255.255.0
默认网关. . . . . : 2025::1
                      192.168.56.3
```


Interconnection(Demo-step2)

2) **Build** a network topology by eNSP, consisting of a “Cloud” device, one PC(PC2), and a Router(R2). **Configure the devices.**

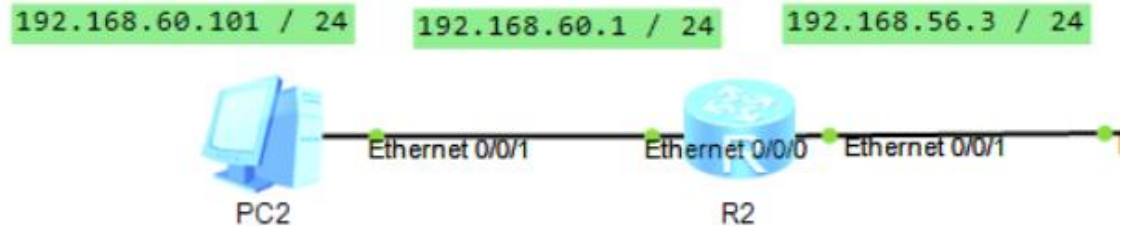
IPv4 Configuration

☒ Static ☐ DHCP

IP Address: 192 . 168 . 60 . 101

Subnet Mask: 255 . 255 . 255 . 0

Gateway: 192 . 168 . 60 . 1



virtual Network in eNSP

R2

```
[Huawei]int eth 0/0/0
[Huawei-Ethernet0/0/0]dis this
#
interface Ethernet0/0/0
 ip address 192.168.60.1 255.255.255.0
#
return
[Huawei-Ethernet0/0/0]int eth 0/0/1
[Huawei-Ethernet0/0/1]dis this
#
interface Ethernet0/0/1
 ip address 192.168.56.3 255.255.255.0
#
return
[Huawei-Ethernet0/0/1]
```

```
[Huawei]dis ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
          Destinations : 6                Routes : 6

Destination/Mask    Proto   Pre  Cost           Flags NextHop         Interface
-----
127.0.0.0/8         Direct  0    0                D    127.0.0.1         InLoopBack0
127.0.0.1/32        Direct  0    0                D    127.0.0.1         InLoopBack0
192.168.56.0/24     Direct  0    0                D    192.168.56.3      Ethernet0/0/1
192.168.56.3/32     Direct  0    0                D    127.0.0.1         Ethernet0/0/1
192.168.60.0/24     Direct  0    0                D    192.168.60.1      Ethernet0/0/0
192.168.60.1/32     Direct  0    0                D    127.0.0.1         Ethernet0/0/0

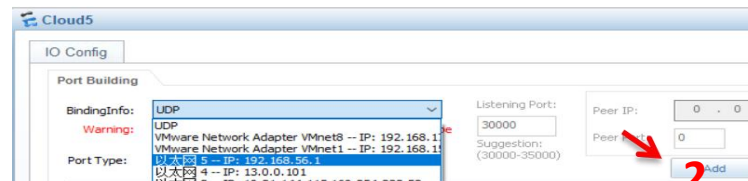
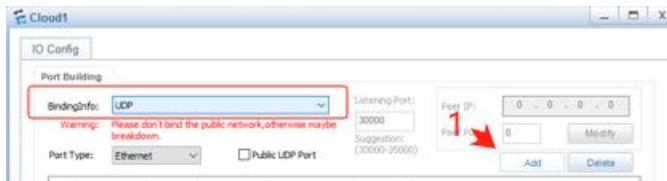
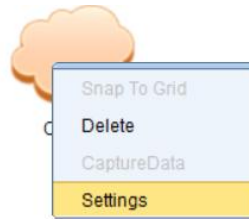
[Huawei]
```

Interconnection(Demo-step3,4)

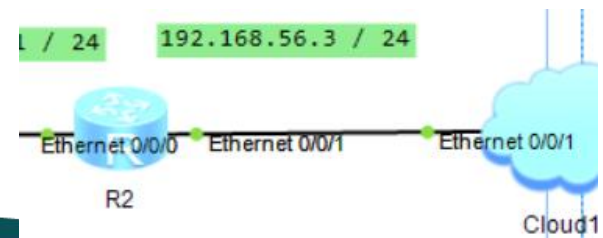
3) Add a “Cloud” device(Cloud1) to the topology, **Configure** the “Cloud1” .

➤ **add** a “Cloud” device to the topology, and **Configure**.

- step0: right click on the “Cloud” , then choose “settings”
- step1,2: add two Bindinfo: (1)UDP (2) network card(whose IP address has been set)
- step3: configure and add Port map settings

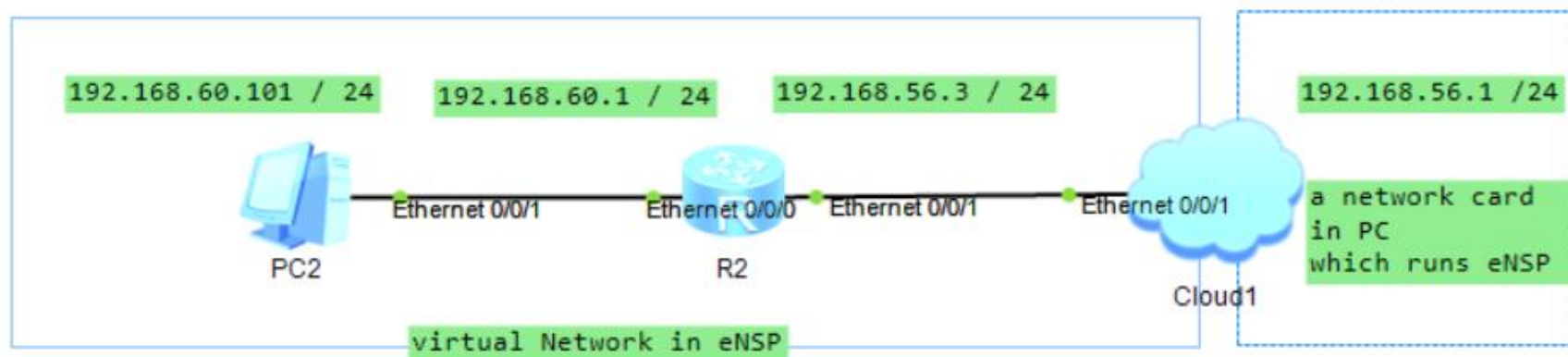


4) **Connect** the “Cloud” device(Cloud1) with the network in the topology.



Interconnection(Demo-step5)

5) **Testing:** invoke “ping” test on PC2 to 192.168.56.1, or invoke “ping” test on 192.168.56.1 to PC2.



Q. While using “ping 192.168.60.101” to do the test, what’s the testing result, explain the reason.

```
PC2
Basic Config  Command  MCPacket  UdpPacket  Console
PC>ping 192.168.56.1

Ping 192.168.56.1: 32 data bytes, Press Ctrl_C to break
From 192.168.56.1: bytes=32 seq=1 ttl=127 time=47 ms
From 192.168.56.1: bytes=32 seq=2 ttl=127 time=31 ms
From 192.168.56.1: bytes=32 seq=3 ttl=127 time=47 ms

--- 192.168.56.1 ping statistics ---
 3 packet(s) transmitted
 3 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 31/41/47 ms

PC>
```

```
C:\Users\sustech>ping 192.168.60.101 -S 192.168.56.1
```

正在 Ping 192.168.60.101 从 192.168.56.1 具有 32 字节的数据: 请求超时。

来自 192.168.60.101 的回复: 字节=32 时间=32ms TTL=127

来自 192.168.60.101 的回复: 字节=32 时间=39ms TTL=127

来自 192.168.60.101 的回复: 字节=32 时间=35ms TTL=127

192.168.60.101 的 Ping 统计信息:

数据包: 已发送 = 4, 已接收 = 3, 丢失 = 1 (25% 丢失),

往返行程的估计时间(以毫秒为单位):

最短 = 32ms, 最长 = 39ms, 平均 = 35ms

```
C:\Users\sustech>
```

Practic3: Cloud tests

- Create a virtual network according to the topology below.
 - The address information in the virtual network is specified by you, and the parameters of the network card are configured according to the network card used on your computer.
 - Make the virtual network in eNSP could communicate with the network card which works on the PC where eNSP is running.
 - Do the testing.

